

Problem

Show how you would use a single op amp to generate

$$v_o = -\int_0^t (v_1 + 4v_2 + 10v_3) d\tau$$

If the integrating capacitor is $C = 5 \mu\text{F}$, obtain the other component values.

Step-by-step solution

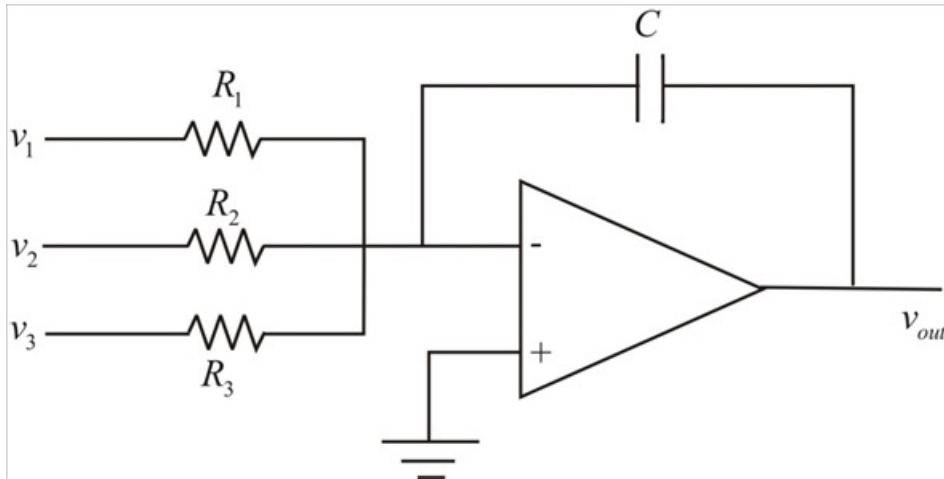
Step 1 of 5

The output voltage of the op amp circuit is

$$v_o = -\int_0^t (v_1 + 4v_2 + 10v_3) dt \dots\dots (1)$$

Here, the output voltage is proportional to the integral of the combination of various input signals.

Draw the circuit by combining a summer with an integrator.



Step 2 of 5

Output voltage equation for the OPAMP circuit is

$$v_o = -\frac{1}{R_1 C} \int_0^t v_1 dt - \frac{1}{R_2 C} \int_0^t v_2 dt - \frac{1}{R_3 C} \int_0^t v_3 dt \dots\dots (2)$$

Compare the two equations (1) and (2)

$$\frac{1}{R_1 C} = 1$$

Substitute $5 \mu\text{F}$ for C .

$$R_1 = \frac{1}{5 \times 10^{-6}}$$

$$R_1 = 200 \text{ k}\Omega$$

Step 3 of 5

$$\frac{1}{R_2 C} = 4$$

Substitute $5 \mu\text{F}$ for C .

$$R_2 = \frac{1}{4 \times 5 \times 10^{-6}}$$

$$R_2 = 50 \text{ k}\Omega$$

Step 4 of 5

$$\frac{1}{R_3 C} = 10$$

Substitute $5 \mu\text{F}$ for C .

$$R_3 = \frac{1}{10 \times 5 \times 10^{-6}}$$

$$R_3 = 20 \text{ k}\Omega$$

Step 5 of 5

So the Resistor values are

$R_1 = 200 \text{ k}\Omega$ $R_2 = 50 \text{ k}\Omega$ $R_3 = 20 \text{ k}\Omega$
